

How AI Makes Decisions

▼ Teacher resources

▼ Printing Options

Dive into the exciting world of artificial intelligence with Code.org's innovative module, "How AI Makes Decisions," tailored for upper elementary school students! This module is named to highlight its focus on unraveling the processes that AI systems use to analyze training data and make decisions. In these five lessons, AI concepts are broken down into engaging, age-appropriate lessons that focus on the critical role of data. Students will use Code.org's AI tools for fun, educational AI experiences while gaining insight into how AI works behind the scenes. Each lesson provides several opportunities for class discussions about AI, bias, and the impact of AI tools on daily life.

New to teaching AI?

Get started with our [self-paced Professional Learning](#).

Week 1	Lesson 1: Making Predictions	Lesson 2: Training AI	Lesson 3: Using AI Bot	Lesson 4: AI Tools in School	Lesson 5: Share Your Voice
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Key	☐ Instructional Lesson	☒ Assessment	☒ Unplugged Lesson
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 Show All Lessons

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▼ Lesson 1: Making Predictions

This lesson will introduce students to the fundamentals of Artificial Intelligence (AI) and machine learning by focusing on predictive data models. Students will get hands-on practice with supervised and reinforcement learning. They'll start with data that has already been sorted and progress to making predictions on their own of how to sort items on screen. In the final level, students will try their hand at sorting data on their own and will enlist a partner to predict how the items have been categorized.

This hands-on experience will be complemented by discussions on the significance of data categorization and the challenges posed by limited datasets. Students will explore a real-world example where AI predictions have been used to help humans. The lesson aims to demystify AI technology, emphasizing its pattern-based logic rather than perceiving it as magical or random.

 1

How is it Sorted?

 2

Sort It Out

 3-5

Sorting Items

3

4

5

 6

Check for Understanding: Matching Level

▼ Lesson 2: Training AI

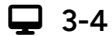
In this lesson, students will learn about how the inputs they give a machine learning model directly impact the predictions that the model makes. First, they will learn about the difference between random predictions and AI predictions, before training their own AI model to make predictions on whether an item is a fish or trash in the ocean. Finally, students will discuss the implications of biased training data by discussing hypothetical scenarios.



Machine Learning Video



Train AI on Fish



Train AI on Sea Creatures

3

4



Teach AI a New Word

▼ Lesson 3: Using AI Bot

In this lesson, students will begin thinking about how to classify real world objects by their features. They will build out two feature vectors (foods and emojis) in order to learn how AI makes decisions based on the probability of an outcome. Students will use "Dance Party: AI Edition" to see first-hand how AI can make tasks more efficient, but that it might come at the cost of customization. Students will consider what it means to collaborate and co-create with AI.



Food Feature Vector



Check For Understanding: Food



Emoji Feature Vector



Check For Understanding: Emojis



Video: How AI Works



Coding with AI

▼ Lesson 4: AI Tools in School

In this lesson, students will evaluate sample student data. They will choose a hypothetical AI tool and consider what training data the tool will need to successfully make a decision. Then, students will write a letter to the school principal explaining their opinion on using AI tools. At the end of the lesson, students will reflect on what they have learned about AI over the course of the 4 lessons.

Students will understand that a predictive AI model is only as good as the information it was trained on. Therefore it is important to train the model on a large amount of data that is relevant to the task at hand. Also, students should recognize that there needs to be human oversight when it comes to working with AI to ensure that any mistakes made by AI get corrected.

☰ 1



Student Survey

☰ 2

Using AI Tools in School (Online Version)

▼ Lesson 5: Share Your Voice

In this lesson, students will create an artifact explaining their opinion on using AI tools. They have the option to write a letter to the principal, make a speech or song, make a poster, create a Sprite Lab project, or code a poem to share their opinion. At the end of the lesson, students will reflect on what they have learned about AI over the course of the module.

👤 1

Choice Level: Share Your Voice

Lesson 1: Making Predictions

45 minutes

Overview

This lesson will introduce students to the fundamentals of Artificial Intelligence (AI) and machine learning by focusing on predictive data models. Students will get hands-on practice with supervised and reinforcement learning. They'll start with data that has already been sorted and progress to making predictions on their own of how to sort items on screen. In the final level, students will try their hand at sorting data on their own and will enlist a partner to predict how the items have been categorized.

This hands-on experience will be complemented by discussions on the significance of data categorization and the challenges posed by limited datasets. Students will explore a real-world example where AI predictions have been used to help humans. The lesson aims to demystify AI technology, emphasizing its pattern-based logic rather than perceiving it as magical or random.

Standards

Full Course Alignment

AI4K12 National Guidelines 2021

- **BI-3** - Computers can learn from data

AI4K12 National Guidelines 2023

- **BI-5** - AI can impact society in both positive and negative ways

CSTA K-12 Computer Science Standards (2017)

- **DA** - Data & Analysis

Agenda

[Optional] Context Setting (10 minutes)

Warm Up (10 minutes)

Vocabulary

Lesson Objectives

Question of the Day

Warm Up

Main Activity (30 minutes)

How Do Computers Learn

Sorting Data

Objectives

Students will be able to:

- Explain different ways to sort the same set of items
- Explain why a computer model might make incorrect predictions
- Store, sort, search, and modify data

Preparation

- Play through the levels on your own
- Preview the sorting activity for students by printing and cutting out the **physical manipulatives**

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- **How AI Makes Predictions** - Slides (**Download**) [▼ Make a Copy](#)
- **[Optional] Physical Sorting Manipulatives** - Resource (**Download**) [▼ Make a Copy](#)

Vocabulary

- **Data** - Information that we collect about the world
- **Feature** - Something that an item is or has. A feature of a triangle is that it has 3 sides.
- **Predict** - To make a guess based on patterns in the data
- **Sort** - To put items into groups based on their features

Turn and Talk

Wrap Up (5 minutes)

Question of the Day

[Optional] Read Aloud Extensions (10 minutes)

- **Training data** - Information that has been sorted, then given to a computer so it can make predictions

Teaching Guide

[Optional] Context Setting (10 minutes)

 **Display:** Show “What is AI?” slide

Do This: As an optional general introduction to AI, show students this short video.

Goal: Some important and relevant points to clarify before beginning this module include: AI stands for Artificial Intelligence. An AI machine is simply a computer that can make decisions after learning from data given to it by humans. A few real world examples of AI include: voice assistants like Siri or Alexa, self driving cars, and robots.

 **Display:** Show “Discuss” slide

Discuss:

- *What is this AI robot doing?*
- *How do you think it learned how to do this?*
- *What do you think would happen if you put a red block on the table?*
- *What are some things this robot probably can not do?*

Discussion Goal: Students should notice that this robot has likely been trained how to sort between blue and green blocks. If it has not been taught how to sort red blocks it might incorrectly sort them into the blue or green items or leave it on the table. This robot likely cannot perform tasks unless they involve human-like motor skills. For example, this robot likely cannot generate text, speak, or see. This is a simple example of AI that directly relates to the activities students will be doing in upcoming lessons.

 Teaching Tip

In this lesson, students will be sorting items by different features. You may want to demonstrate how to do this by using physical manipulatives before students complete the levels online. You can use items that you have in your classroom or print and cut out the **sorting manipulatives** that students will see in the following levels.

Warm Up (10 minutes)

Vocabulary

 **Display:** Show “Vocabulary” slide

Say:

- **Data:** Information that we collect about the world.
- **Sort:** To put items into groups based on their features.
- **Feature:** Something that an item is or has. A feature of a triangle is that it has 3 sides.
- **Predict:** To make a guess based on patterns in the data.
- **Training data:** Information that has been sorted, then given to a computer so it can make predictions

Lesson Objectives

 **Display:** Show “Lesson Objectives” slide

Say: By the end of this lesson, you will be able to:

- *Explain different ways to sort the same set of items*
- *Explain how having more data can lead to more accurate predictions*

Question of the Day

 **Display:** Show “Question of the Day” slide

Say: *What features help us decide how to sort a group of objects?*

Warm Up

 **Display:** Show “Warm Up” slide

Discuss: Which of these doesn’t belong? Why? Explain what feature makes it different.

Do This: Have students discuss different ways of grouping these items in groups of 3 and to explain their reasoning. Some examples are by: shape, color, photo orientation, and toppings. You can also get students thinking like a computer. Ask them “How would a computer tell the difference between these four items? What would it need to know about each pastry?”

Discussion Goal: Students should understand that there are different ways of grouping and sorting the same items based on their features. Challenge them to come up with as many different ways of sorting these 4 items as they can.

Main Activity (30 minutes)

How Do Computers Learn (3 minutes)

 **Display:** Show “How do computers learn?” slide

Do This: Click through the diagram, step-by-step, reading each step aloud to students.

Say: Today you are going to act as the computer. Someone has already sorted the items behind the scenes. You have to predict what the pattern is for each level and sort the items into their correct box.

Sorting Data (25 minutes)

 **Display:** Show “Levels 1-6: Predict the Pattern” slide

Do This: Transition students to their devices. Have them go to Level 1 and answer the multiple choice question.

Goal: In Levels 1-6, students get to act as if they were AI, and make predictions about where the items should go. Levels 1 and 2 are examples of supervised machine learning. The inputs have been labeled, by color and by wings, which means that students will sort the items according to their label. Levels 3-5 are examples of reinforcement learning. The data is unlabeled and students have to classify it by receiving feedback on whether or not their prediction was correct.



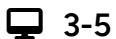
How is it Sorted?

Teaching Tip

Before allowing students on their devices, consider modeling this level for the class. If students have never encountered a multiple choice question on the Code.org platform before they may be unfamiliar with how it works. On Level 1, instruct students to read and answer the question before clicking Run. In the following levels, they will need to click Run in order to drag items around on screen.



Sort It Out



Sorting Items

3

4

5



Check for Understanding: Matching Level

Display: Show "Level 7: Your Turn!" slide

Do This: Once students have finished Levels 1-6, have them choose one of the sublevels in Level 7 Choice Level and pair them up with a partner. One student will sort the items in the level while the other partner has to guess the pattern. Then have them switch roles.



Your Turn!

Turn and Talk (2 minutes)

Display: Show "Turn and Talk" slide

Discuss: Ask students "Which level was the easiest for you to sort? Why?"

Discussion Goal: Have students reflect on patterns they noticed in the data. They should be able to articulate that having more sorted data leads to better, more accurate predictions. A level in which items are not initially sorted becomes easier to sort as students begin to place items in the correct boxes.

Wrap Up (5 minutes)

Question of the Day

Display: Show "Question of the Day" slide

Discuss: What features help us decide how to sort a group of objects?

Discussion Goal: Students should recognize that it depends on the items. For example, they can't always group items by shape if all the items are the same shape. Students should first figure out what features these items have in common and then determine how to sort them into multiple groups by a shared feature.

If students need an example, choose one of the options in Lesson 1, Level 7. Ask students how they might sort the items on screen. What features do they notice the items have in common?

 Teaching Tip

If you need to give students an assessment at the end of each lesson, use the Question of the Day as an assessment of student learning, e.g. an exit ticket.

 **Display:** Show "Reflect" slide

Discuss: Here are two real world examples that use AI to sort data. Can you think of more?

Discussion Goal: Discuss the two examples in this slide. [Optional: Show students the short (1 min) video about BakeryScan AI]. Then, see if students can come up with additional examples of how a sorting algorithm could be used in the real world.

Some hypothetical examples: sorting between dogs and cats, laundry sorter that sorts pants/shirts/socks, sorting out M&Ms from the rest of the trail mix.

Some real world examples: self-driving cars detecting empty crosswalks vs. a pedestrian crossing, sorting by trash/compost/recycling, Roomba vacuums determining if there is an obstacle or not, detecting if an email/text/call is spam or not.

Here are some additional news examples:

- **Food Waste Is a Serious Problem. AI Is Trying to Solve It**
- **AI system used to create a patrol route for conservationists to help prevent poaching attacks**
- **AI tech can identify genetic disorders from a person's face**
- **MIT AI tool can predict breast cancer up to 5 years early**

[Optional] Read Aloud Extensions (10 minutes)

 **Display:** Show "Read Aloud Extensions" slide

Do This: Here are two additional read aloud extensions to show students.

- **Are You My Mother?** by P.D. Eastman
- **Grandma's Button Box** by Linda Aber

While these books were not designed for teaching about machine learning or sorting algorithms, their content can be used to start a discussion about features that the different characters or buttons have. Encourage students to start noticing patterns around them in the classroom and beyond where items might be sorted by their features.



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Lesson 2: Training AI

45 minutes

Overview

In this lesson, students will learn about how the inputs they give a machine learning model directly impact the predictions that the model makes. First, they will learn about the difference between random predictions and AI predictions, before training their own AI model to make predictions on whether an item is a fish or trash in the ocean. Finally, students will discuss the implications of biased training data by discussing hypothetical scenarios.

Standards

Full Course Alignment

AI4K12 National Guidelines 2021

- **BI-3** - Computers can learn from data

AI4K12 National Guidelines 2023

- **BI-5** - AI can impact society in both positive and negative ways

CSTA K-12 Computer Science Standards (2017)

- **DA** - Data & Analysis

Agenda

Warm Up (10 minutes)

Vocabulary

Lesson Objectives

Question of the Day

Warm Up

Main Activity (30 minutes)

AI for Oceans

Turn and Talk

Reflect

Wrap Up (5 minutes)

Question of the Day

Wrap Up

Objectives

Students will be able to:

- Explain how training a model can look different based on someone's opinion
- Explain the difference between a random prediction and an AI prediction
- Train a machine learning algorithm on chosen features

Preparation

- Play through the levels on your own

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- **How AI Makes Predictions** - Slides
([Download](#)) [▼ Make a Copy](#)

Vocabulary

- **Accurate** - Another way to say "correct"
- **Artificial Intelligence (A.I.)** - A way for computers to make decisions by themselves
- **Bias** - Favoring some things but not others, which can lead to an unfair decision
- **Data** - Information that we collect about the world
- **Machine Learning** - How computers recognize patterns and make decisions without being explicitly programmed

- **Predict** - To make a guess based on patterns in the data
- **Training** - Teaching a computer to do tasks by showing it many examples
- **Training data** - Information that has been sorted, then given to a computer so it can make predictions

Teaching Guide

Warm Up (10 minutes)

Vocabulary

 **Display:** Show "New Vocabulary" slide

Say:

- **Machine Learning** - How computers recognize patterns and make decisions
- **Artificial Intelligence** - A way for computers to make decisions by themselves
- **Accurate** - Another way to say "correct"
- **Bias** - Favoring some things but not others, which can lead to an unfair decision

Lesson Objectives

 **Display:** Show "Lesson Objectives" slide

Say: By the end of this lesson, you will be able to:

- *Explain the difference between a random prediction and an AI prediction*
- *Train a machine learning algorithm on chosen features*
- *Explain how training a model can look different based on someone's opinion*

Question of the Day

 **Display:** Show "Question of the Day" slide

Say: *How do computers become more accurate at making predictions?*

Warm Up

 **Display:** Show "Warm Up" slide

Do This: Generate a selection of random books using the Random Books Generator site. Show students the 6 books that were generated, and explain that these recommendations are randomly selected from an existing list of books. Then, search for a student's favorite book on Amazon and scroll down to look at Items to "Explore." For example, check out "More items" to explore on this search for *Diary of a Wimpy Kid*. If students aren't familiar with Amazon, ask them what types of videos are recommended for them when they finish watching a video on YouTube.

Discuss:

- How did each site decide which books to feature?
- Which set of books are you more likely to read?
- How do you think these book recommendations might be different from ones given to your friend, parent, teacher, or librarian?

Discussion Goal: Students should be able to explain that the Random Book Generator site generates books randomly from an unsorted/untagged list of books. If you were to keep clicking “Generate,” it would pick 6 books randomly from the list each time.

Context: Amazon collects data associated with each book. For example, it tracks what other books were bought by people who bought the book that the user searched for. It also uses a recommendation algorithm to look for similar books (author, grade level, content) to the book being searched for. Searching for *Diary of a Wimpy Kid* on Amazon results in recommendations for other books in the Wimpy Kid series.

Students will likely find books in the random book generator that are not at the right reading level for them or that they don’t find interesting. In comparison, Amazon might recommend more familiar or relevant books. They should understand that if their parent/friend/teacher/librarian clicks “Generate” on the random book generator they are equally likely to get the same book recommendations. However, if their parent/friend/teacher/librarian were to type in their own favorite book into Amazon’s search, they would get more customized recommendations.

Teaching Tip

If you’re unable to access the Random Book Generator site or Amazon on your school’s WiFi network, show the following slide in place of navigating to those websites.

Main Activity (30 minutes)

AI for Oceans (20 minutes)

 **Display:** Show “Machine Learning” video

Discuss: What does the speaker mean when she says computers learn through “trial and error” and “lots of practice”?

Discussion Goal: Have students reflect on the previous lesson. Sometimes they made a guess about what box to put an item in and they were incorrect. They received feedback, then corrected their guess, and moved on to make a prediction about the next item. This is an example of “trial and error” or reinforcement learning. A machine learning model gets fed data, makes predictions, and then gets feedback based on how accurate those predictions were in order to make better predictions in the future.

Discuss: How do you think this Bot is able to tell the difference between cars and bikes?

Discussion Goal: Students should be able to explain that this bot was trained on many different images which were tagged as either a bicycle or a car. Some features that it might be looking for: 2 or 4 wheels, how much of the background is visible behind the image, windows or no windows, handlebars or no handlebars.



Machine Learning Video

 **Display:** Show “Level 2” slide

Do this: Transition students to their devices. Have them go to Level 2 and sort the items, training the algorithm to distinguish between fish and trash. Have students keep track of the counter in the top right corner to see how many images they have trained AI on. See if they can guess how many images AI will need in order to make accurate predictions.

Teaching Tip

Depending on how ready your students are for independent online activities and how comfortable they are reading the text on screen, you may want to complete this level as a class. Display this level for the whole class to see. As you train AI Bot, have the whole class give a thumbs up or thumbs down to indicate whether the model should recognize the item as a fish or not.

 **Display:** Show “Levels 3-4” slide

Have students complete Levels 3 and 4 on their own devices. Level 3 introduces the idea that “AI only learns what we teach it.” Students should be aware that we want to train our model to recognize all creatures that belong in the water. In Level 4, students will retrain the model to recognize fish, otters, dolphins, and other sea creatures.

 **Display:** Show “Level 5” slide

Have students complete Level 5 by choosing a word to teach A.I. Group students by the word they chose and have them compare their results. Students should start to notice that even though they chose the same word as a classmate, they may have labeled the fish training data differently and, as a result, their AI made different predictions about which fish to include.

Turn and Talk (5 minutes)

 **Display:** Show “Turn and Talk” slide

Discuss:

- *Why might your AI predictions look different from your classmate’s AI predictions even though you trained AI on the same word?*
- *What features were most important to you when you were labeling fish?*

Discussion Goal: Students should be able to explain that their training dataset reflects their own opinions. Students might have different opinions about which fish looks “Fast” or “Happy” because these words are subjective, meaning they could be different for each person. Additionally, students might have chosen to look at certain features of a fish and not others, such as if it was showing its teeth or if its eyebrows were raised. AI makes predictions based on its training data, so students with different training datasets will likely notice different AI predictions.

Reflect (5 minutes)

 **Display:** Show “Reflect” slide

Discuss:

- What would happen if all the fish labeled 'Angry' were blue and all the fish labeled 'Awesome' were red?
- What issues might that cause in the long run?

Discussion Goal: If students were to label only blue fish as “angry” and only red fish as “awesome”, the computer would learn that biased pattern and might start labeling all blue fish as “angry” or all the red fish as “awesome” even when they’re not. These inaccurate and unfair predictions, also called bias, could cause issues in the future if the computer learns something untrue about a particular type of fish. We can help solve this problem, by giving the computer training data collected from a variety of sources and making sure the data represents all types of fish.

Wrap Up (5 minutes)

Question of the Day

 **Display:** Show “Question of the Day” slide

Discuss: *How do computers become more accurate at making predictions?*

Discussion Goal: Students should understand that the more data they give AI, the more accurate its predictions are: They should understand that tagging data can be subjective. For example, students might have different opinions about which fish look “silly”. Getting data from all the students in the class helps eliminate bias when using training data that might be based on opinions. Students should consider multiple scenarios and not make any assumptions about what AI knows. For example, AI doesn’t know that otters and dolphins belong in the ocean if it was only trained on fish. Students will get more practice doing this in the Wrap Up question.

Wrap Up

 **Display:** Show “Wrap Up” slide

Pretend your school is using Attendance Bot, an AI device that scans the classroom and checks if a student is present or absent.

If a student is sitting in their seat at 9:00 am, they are marked Present. Otherwise, they are marked Absent and their teacher, principal, and parent is notified.

- What are the pros and cons of using Attendance Bot?
- What could cause Attendance Bot to mark a Present student as Absent?
- What should the makers of Attendance Bot consider when training their AI?

Discussion Goal: Have students consider that this AI device may not always be more accurate than a teacher taking attendance because it can only see the class at one specific moment in time (con). On the flip side, it might make a teacher’s life easier because it automates a task that a teacher would have to do by hand (pro). Students might also consider how the AI tool can’t ask questions or see nuanced scenarios, such as a student who is present but not in their seat. Have students consider examples of when Attendance Bot might be wrong where a human would not be.

Examples of when Attendance Bot might mark a present student as absent:

- If the student is present but somewhere else in the classroom, e.g. at the teacher’s desk, passing out papers, in another seat
- If the student is kneeling in their chair or bending down to pick something off the floor
- If a student is in the bathroom
- If the student is in a wheelchair or uses a mobility device that prevents them from being able to sit in the school chair

The makers of Attendance Bot should consider the different scenarios above when training their AI. They might also want to have the Bot take attendance throughout the day to make sure that it is more accurate.



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Lesson 3: Using AI Bot

45 minutes

Overview

In this lesson, students will begin thinking about how to classify real world objects by their features. They will build out two feature vectors (foods and emojis) in order to learn how AI makes decisions based on the probability of an outcome. Students will use "Dance Party: AI Edition" to see first-hand how AI can make tasks more efficient, but that it might come at the cost of customization. Students will consider what it means to collaborate and co-create with AI.

Standards

Full Course Alignment

AI4K12 National Guidelines 2021

- **BI-3** - Computers can learn from data

AI4K12 National Guidelines 2023

- **BI-2** - Agents maintain representations of the world and use them for reasoning
- **BI-5** - AI can impact society in both positive and negative ways

Agenda

Warm Up (10 minutes)

Vocabulary

Lesson Objectives

Question of the Day

Warm Up

Main Activity (30 minutes)

Introduction to AI Dance Party

Food Feature Vector

Independent Work Time

AI Dance Party

Reflect

[Optional] Additional Context Slides

Wrap Up (5 minutes)

Question of the Day

Wrap Up

Objectives

Students will be able to:

- Build out your own feature vector
- Experience how AI can make tasks more efficient
- Explain why AI may choose a different outcome than you

Preparation

- Play through the levels on your own

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- **How AI Makes Predictions** - Slides ([Download](#)) [▼ Make a Copy](#)

Vocabulary

- **Artificial Intelligence (A.I.)** - A way for computers to make decisions by themselves
- **Bias** - Favoring some things but not others, which can lead to an unfair decision
- **Data** - Information that we collect about the world
- **Feature Vector** - Assigning a number value to certain features of an object so that a machine can find patterns between multiple objects in the data

Teaching Guide

Warm Up (10 minutes)

Vocabulary

 **Display:** Show “New Vocabulary” slide

Say:

- **Feature Vector** - Assigning a number value to certain features of an object so that a machine can find patterns between multiple objects in the data

Lesson Objectives

 **Display:** Show “Lesson Objectives” slide

Say: By the end of this lesson, you will be able to:

- *Build out your own feature vector*
- *Experience how AI can make tasks more efficient*
- *Explain why AI may make different predictions when asked the same question*

Question of the Day

 **Display:** Show “Question of the Day” slide

Say: Why might AI give different answers to the same question? For example, I asked AI what colors are in a sunset. It told me, “Orange and red”. When my classmate asked the AI, it said, “Purple and pink.”

Warm Up

 **Display:** Show “Warm Up” slide

Do This: Read the slide aloud.

 **Display:** Show “Preparing the data for AI” slide

Do This: Help students look at each animal’s number line. Ask if they can figure out how many employees were surveyed. Point out that each animal was rated by each employee on a scale from 1-10 in terms of how cute it is and its size.

 **Display:** Show “How do computers learn?” slide

Do This: Click through the diagram, step-by-step, reading each step aloud to students.

 **Display:** Show “A.I.’s Training Data” slide

Say: Where each animal has been placed on the graph represents the average, or “middle”, of all pet shop employees' opinions of that animal. We can see that all the pet store employees agreed that lizards and rats are small and horses are big. Most employees thought that puppies and rats were cute.

Discuss: *Why did AI pet picker choose a rat? What else might AI pet picker choose if you asked for a house pet again?*

Discussion Goal: AI pet picker chose a rat because it is in the “house pet” section of the chart along with the other “small” and “cute” animals. If asked again, AI pet picker might respond with “puppy” or even “lizard”.

 **Display:** Show “[Optional] Extension: Class Results for favorite pet” slide

 Teaching Tip

This slide can be used to reinforce the idea of how AI makes predictions using probabilities. For a given prompt, AI might have several options to select from. It will usually choose the most popular option (the option with the highest probability) but can occasionally choose options with lower probabilities.

Main Activity (30 minutes)

Introduction to AI Dance Party (3 minutes)

 **Display:** Show “Dance Party: AI Edition - How AI Works” video

Discuss: What information was this AI Bot trained on?

Discussion Goal: AI is learning to recognize different emojis and associate words with them. It’s using information taken from the internet to determine what words fit with each emoji.

Discuss: What 3 emojis did the user choose?

Discussion Goal: This is a check for understanding to make sure students understand how they will select 3 different emojis for AI Bot.

 **Display:** Show “How does AI make associations?” slide

Say: AI only knows the information that we give it. Before we use AI Bot in Dance Party, let’s teach AI by making some associations of our own:

1. First you’ll make a feature vector of food by its **sweetness** and how much you **like to eat it**
2. Then you’ll make a feature vector of emojis by how **colorful** and **happy** they are.

Food Feature Vector (5 minutes)

 **Display:** Show “Level 1: Food Feature Vector” slide

Do this: Do This: Transition students to their devices. Have them go to Level 1. Tell students to click the orange Run button under the playspace and choose a food item at the top of the screen. Have students store each food item in their feature vector by clicking and dragging it into the box.

First, have students move the item along the horizontal “Sweetness” axis, from Not Sweet to Really Sweet. Then, have students move the item up the vertical “Like to eat” axis depending on how much they like the food, from Not for me! To Love it!



Food Feature Vector

 Teaching Tip

Depending on how ready your students are for plotting items on a feature vector, you may want to complete this level as a class first. Display this level for the whole class to see. If students disagree with where you’ve placed an item, remind them that this level is based on individual opinion and they should reflect their own opinions in their answers.

Independent Work Time (10 minutes)

 **Display:** Show "Level 2" slide

Do this: Have students complete Levels 2-5 independently. In Level 5, students have the option to re-watch the "Dance Party: AI Edition - How AI Works" video on their own.

 2

Check For Understanding: Food

 3

Emoji Feature Vector

 4

Check For Understanding: Emojis

AI Dance Party (10 minutes)

 **Display:** Show "Picking a Dance Party Stage" slide

Do This: Read the slide aloud for students. This slide is designed to set up how AI can help us be more efficient with our tasks.

 **Display:** Show "AI is here to help!" slide

Say: In Level 6, you will use a block to make AI Bot generate foreground and background effects for your dance. You choose 3 emojis and AI Bot will choose the effects that best match those emojis.

 **Display:** Show "How does AI choose effects?" slides

Do This: Read the slides aloud. Have students choose the three palettes that they believe best represent the smiley face emoji or the words "happy", "sunny", and "bright". Then compare their choices to AI's choices on the next slide.

 5

Video: How AI Works

Do This: This is the same video that shows up in the slides at the beginning of the Main Activity. If you haven't already, play the video for students or give students the option to watch it on their own.

 **Display:** Show "Level 6: Coding with AI" slide

Do This: Open Level 6 on your screen and display it to the whole class. Model how the AI block works: Choose one of the songs from the level and click Play. Survey students to see what emojis they would choose based on the song. Point out how choosing an emoji to match the mood of the song is subjective and students can have different answers. Based on which 3 emojis are chosen, AI Bot might recommend different background effects for the same song.

Click "Generate" and see what effects get chosen. Explain that if students don't like the effects, they can Regenerate effects using the same 3 emojis, Use the code and update the dropdown fields themselves, or Start over with 3 new emojis.

 6

Coding with AI

Reflect (2 minutes)

 **Display:** Show “Do you prefer selecting the effects or having AI select the effects for your dancer’s stage? Why?” slide

Discussion Goal: Have students reflect on the pros and cons of each option. By selecting the effects themselves, they can create any combination they want but it’s often tedious to click each option from the dropdown menu. Having AI select the effects based on 3 emojis is easier, but may not yield the exact results that students want. One potential solution is for students to work with AI by having it create a first draft and having students edit the results. In this case, students can have AI generate the effects and then they can click “Use code” and make any adjustments.

[Optional] Additional Context Slides

 **Display:** Show “Additional Context” slides These slides are optional. They provide a more detailed explanation of how AI Bot makes its decisions. You may want to encourage students to click on the bar graph to see how AI Bot chose effects based on their emoji selection.

Wrap Up (5 minutes)

Question of the Day

 **Display:** Show “Question of the Day” slide

Discuss: *Why might AI give different answers to the same question?*

For example, I asked AI what colors are in a sunset. It told me, “Orange and red”. When my classmate asked the AI it said, “Purple and pink”.

Discussion Goal: Make sure students understand that AI is trained on billions of data points representing many different opinions. Since not everyone agrees on for example, what colors are in a sunset, AI will make a prediction based on the probabilities of different responses. If most people say that a sunset contains the colors orange and red, then AI will usually say “orange and red”. But since sometimes people see purple and pink in a sunset, AI will occasionally say “purple and pink.” Just as asking different people what colors are present in a sunset will yield different answers, asking AI the same question may result in different responses.

Have students reflect on the Warm Up activity in which AI Pet Picker chose a rat, when we were expecting it to choose a puppy. Point out how each Pet Shop employee had different opinions about how cute each animal is and how big it is. Sometimes conflicting opinions in the training data can lead to conflicting AI predictions. If AI Pet Picker was trained on the opinions of everyone in the world, it might not pick the pet that you or I had in mind, but it would certainly reflect someone’s opinion from the training data.

Additionally, you can point out that there can be small and large effects of AI making a different or even incorrect prediction. In the previous lesson, if AI couldn’t correctly predict that a starfish belonged in the ocean, it would assume that it was trash which needed to be removed. In the real world, this could cause large negative environmental impacts.

Wrap Up

 **Display:** Show “Wrap Up” slide

Discuss: *Choose one of the following jobs: Veterinarian, Teacher, Bus Driver, Artist.*

Now, imagine if the person doing this job were replaced by an AI robot.

- *Would you want AI to replace a human in this job? Why or why not?*

- *What would happen if this AI got something wrong?*

Discussion Goal: Students should be able to give examples of what might happen if this AI got something wrong: misdiagnosis, misinformation, taking a wrong turn, etc. The takeaway here should be that AI can help us automate tasks so we can do them faster, e.g. pick out background effects, but AI needs oversight. There should be a partnership between humans and AI so that we are not entrusting AI to make the final decision or to operate unchecked.



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Lesson 4: AI Tools in School

45 minutes

Overview

In this lesson, students will evaluate sample student data. They will choose a hypothetical AI tool and consider what training data the tool will need to successfully make a decision. Then, students will write a letter to the school principal explaining their opinion on using AI tools. At the end of the lesson, students will reflect on what they have learned about AI over the course of the 4 lessons.

Students will understand that a predictive AI model is only as good as the information it was trained on. Therefore it is important to train the model on a large amount of data that is relevant to the task at hand. Also, students should recognize that there needs to be human oversight when it comes to working with AI to ensure that any mistakes made by AI get corrected.

Standards

Full Course Alignment

AI4K12 National Guidelines 2021

- **BI-3** - Computers can learn from data

AI4K12 National Guidelines 2023

- **BI-5** - AI can impact society in both positive and negative ways

Agenda

Warm Up (10 minutes)

Vocabulary

Lesson Objectives

Question of the Day

Warm Up

Main Activity (30 minutes)

How Do Computers Learn

Using AI Tools In School

Wrap Up (5 minutes)

Question of the Day

Wrap Up

Objectives

Students will be able to:

- Consider the pros and cons of hypothetical (made up) AI tools
- Look at a student data and notice patterns
- Write a letter to your school principal explaining why or why not the school should use AI tools to make decisions

Preparation

- Decide in advance if you want students to do the Warm Up and Main Activity on their devices or as an unplugged activity
- If Unplugged, print out the **Student Survey** and **Activity Guide** in advance. Make sure students have a way to access the **Student Data** resource.
- If using devices, make sure you are able to access Student Survey results in your Teacher Dashboard.

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- **How AI Makes Predictions** - Slides (**Download**) ▼ Make a Copy
- **[KEY] Using AI Tools in School** - Answer Key (**Download**) ▼ Make a Copy

For the students

- **Student Data** - Resource (**Download**)

- **Using AI Tools in School** - Activity Guide ([Download](#)) [▼ Make a Copy](#)
- **[Optional] Student Survey Activity** - Activity Guide ([Download](#)) [▼ Make a Copy](#)

Vocabulary

- **Artificial Intelligence (A.I.)** - A way for computers to make decisions by themselves
- **Data** - Information that we collect about the world
- **Training** - Teaching a computer to do tasks by showing it many examples
- **Training data** - Information that has been sorted, then given to a computer so it can make predictions

Teaching Guide

Warm Up (10 minutes)

Vocabulary

 **Display:** Show "Vocabulary" slide

Say:

- **Artificial Intelligence** - A way for computers to make decisions by themselves
- **Data** - Information that we collect about the world.
- **Training** - Teaching a computer to do tasks by showing it many examples.
- **Training data** - Information that has been sorted, then given to a computer so it can make predictions

Lesson Objectives

 **Display:** Show "Lesson Objectives" slide

Say: By the end of this lesson, you will be able to:

- *Read a sample student dataset and notice patterns*
- *Explain what data makes sense to give an AI tool that is making a decision in school*

Question of the Day

 **Display:** Show "Question of the Day" slide

Say: *How can you prevent bias in your training data?*

Warm Up (10 minutes)

 **Display:** Show “Warm Up” slide

Teaching Tip

The following slides present two options for completing this warm up. You can either have students answer the questions independently in Level 1 (Online Version) or survey each other in person and collect the data on paper (Paper Version). For both options, you have the choice to use sample student data if you don't have enough time for students to provide or collect data on their own.

Say: We're going to collect some data of our own. In a minute, you will go to Level 1 and answer the four multiple choice questions about your favorite ice cream flavor, pizza topping, game, and color. If your favorite answer doesn't appear, choose E. Something Else. Then we'll look at the class data altogether.

Discuss: Project your class responses on the board. Ask students to look at the data and answer the following four questions:

1. What does each row represent?
2. What information is captured in each column?
3. What decisions could you make using this dataset?
4. Would you be able to make a decision for the whole school using just this data?

Discussion Goal: Students should understand that each row represents the results of one survey question. Each column represents how many students chose that response (represented as a percentage of the entire class). Some decisions that could be made based on this dataset: what flavors of ice cream or pizza to buy for a class party, whether recess should be inside or outside, what colors to use for classroom decorations or class folders. This data shouldn't be used to make a decision on behalf of the entire school because it's biased towards the opinions of one classroom. Students should keep in mind any bias that could result from a dataset having limited or subjective data.

Main Activity (30 minutes)

How Do Computers Learn (3 minutes)

 **Display:** Show “How do computers learn?” slide

Do This: Click through the diagram, step-by-step, reading each step aloud to students.

Say: Today you are going to act as the human training the computer. It is up to you to decide what information to give the computer (and what information to leave out) so that it can make the best predictions possible.

 1



Student Survey

Using AI Tools In School (27 minutes)

 **Display:** Show “Using AI Tools in School” slide

 2

Using AI Tools in School (Online Version)

The following slides present two options for completing this student activity. You can either have students answer the questions independently in Level 2 (Online Version) or go to the next slide **Using AI Tools in School** handout (Paper Version).

Say: Now it's your turn to investigate ways AI can help us in the real world. You'll look at a few examples of AI tools that can be used in school and decide which one you want to use.

Once you've picked your AI tool, you're going to look at some student data. You'll think about what data will be most important to give to AI, what data you don't have that you would add to the dataset, and what data you would remove from the dataset.

Do This: Direct students to get started on Level 2.

You can preview the link for students **here** and point out the different rows and columns. The dataset can also be found on the last page of the Using AI Tools in School paper handout.

Wrap Up (5 minutes)

Question of the Day

Display: Show "Question of the Day" slide

Discuss: *How can you prevent bias in your training data?*

Discussion Goal: Students should understand that training data can be subjective, for example what an "angry" fish looks like or someone's favorite ice cream flavor. It is important to consider what data should and should not be collected. When having AI decide where to seat people in the cafeteria, including what languages they speak is very helpful. On the flip side, certain data which may seem irrelevant, such as the number of detentions a student received, and could even lead AI to make a biased decision. If this is the case, this data should likely be left out of the training dataset and/or AI's predictions should be evaluated by a human.

Wrap Up (5 minutes)

Display: Show "Wrap Up" slide

Discuss: *AI Activity Picker recommends activities for kids but it keeps recommending only beach activities and other outdoor activities.*

- *Why is that a problem?*
- *What data can we collect to solve the problem?*

Discussion Goal: Students should understand that this AI shows a bias towards beach and outdoor activities due to the data it was provided. It is a problem because kids might be looking for indoor activities and / or not live near the beach. In order to make this AI tool useful for all kids, we should train it on more data that includes indoor activities and activities that aren't location-specific.



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Lesson 5: Share Your Voice

45 minutes

Overview

In this lesson, students will create an artifact explaining their opinion on using AI tools. They have the option to write a letter to the principal, make a speech or song, make a poster, create a Sprite Lab project, or code a poem to share their opinion. At the end of the lesson, students will reflect on what they have learned about AI over the course of the module.

Standards

Full Course Alignment

AI4K12 National Guidelines 2021

- **BI-3** - Computers can learn from data

AI4K12 National Guidelines 2023

- **BI-5** - AI can impact society in both positive and negative ways

Agenda

Warm Up (10 minutes)

Vocabulary

Lesson Objectives

Question of the Day

Warm Up

Main Activity (30 minutes)

Share Your Voice

Wrap Up (5 minutes)

Reflection

Question of the Day

Wrap Up

Objectives

Students will be able to:

- Consider the pros and cons of AI tools
- Explain their opinion on AI tools to their classmates and teachers

Preparation

- Survey the different level types in the **choice level** to differentiate options for students
- Prepare any materials students will need to write a letter, design a poster, or complete their projects

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- **How AI Makes Predictions** - Slides (**Download**) ▼ Make a Copy

For the students

- **Share Your Voice - AI Poster** - Handout (**Download**) ▼ Make a Copy

Vocabulary

- **Artificial Intelligence (A.I.)** - A way for computers to make decisions by themselves
- **Bias** - Favoring some things but not others, which can lead to an unfair decision

- **Data** - Information that we collect about the world
- **Training** - Teaching a computer to do tasks by showing it many examples

Teaching Guide

Warm Up (10 minutes)

Vocabulary

 **Display:** Show "Vocabulary" slide

Say:

- **Artificial Intelligence** - A way for computers to make decisions by themselves
- **Data** - Information that we collect about the world.
- **Training** - Teaching a computer to do tasks by showing it many examples.
- **Bias** - When a decision favors some things and not others

Lesson Objectives

 **Display:** Show "Lesson Objectives" slide

Say: By the end of this lesson, you will be able to:

- *Consider the pros and cons of AI tools*
- *Explain your opinion on AI tools to your classmates and teachers*

Question of the Day

 **Display:** Show "Question of the Day" slide

Say: *How do we decide if a task should use AI or not?*

Warm Up

 **Display:** Show "Warm Up" slide

Discuss: Consider one of the AI tools from the last lesson that chose:

- Seats in the cafeteria
- Teams for gym class
- Student of the Month

What are the pros and cons of using AI tools to make these decisions, instead of letting a human decide?

Discussion Goal: Have students think about the different AI tools from the previous lesson. While using AI could save time, it may lead to outcomes that are undesirable, such as placing students who don't get along next to each other in the cafeteria. Also, AI might make decisions that a human wouldn't make. Depending on the situation this may or may not be a positive outcome.

Main Activity (30 minutes)

Share Your Voice (30 minutes)

 **Display:** Show “Level 1: Share Your Voice” slide

Say: Today, you'll get a chance to share your opinion on AI by writing a letter to your principal, creating a poster, or expressing yourself in another way. You'll share whether or not you think AI tools should be used to help make a decision in school. Take a minute to decide which of these options you would like to choose.

The number of chili peppers next to the option shows how difficult it is. So make sure you choose an option that you'll be able to finish in class today. When you're ready, go to Level 1 and get started.

Do This: Have students choose from one of the following options:

- Write a letter to the principal
- Make a speech, song, or rap
- Design a poster
- Create a project in Sprite Lab
- Code a poem

 1

Choice Level: Share Your Voice

Wrap Up (5 minutes)

Reflection

 **Display:** Show “Reflection” slide

Question of the Day

 **Display:** Show “Question of the Day” slide

Discuss: *How do we decide if a task should use AI or not?*

Discussion Goal: Have students think about the different examples that they learned about in the previous lessons: BakeryScan and machine learning sorting algorithms, Amazon's recommendation algorithm, AI for Oceans, AI Bot to pick background effects, and hypothetical scenarios involving AI tools. They should understand that there are pros and cons to using AI tools and the decision is not “one size fits all”. Even in scenarios where they determine it makes sense to use AI, there should still be human involvement in making sure the outcome is fair and correct.

Wrap Up

 **Display:** Show “Glows, Grows, Want-to-Knows” slide

Discuss: Have students discuss their letters, posters, and songs they made today and have them reflect on something they learned about AI in the previous lessons.

Discussion Goal: Students should share what they have learned and any questions they still have about AI. This is an ongoing opportunity for students to continue learning and developing opinions about AI. Encourage students to keep coming up with questions they have about how AI works and where it shows up around them.



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